UCONN
COLLEGE OF ENGINEERING
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UConn Quant: Team 53

A momentum based trading algorithm

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Introduction and Motivation

The S&P 500 is a widely-used benchmark, but passive index investing leaves investors fully exposed to major market drawdowns. During the global financial crisis, the index fell by about 57% from October 2007 to March 2009, and during the Covid-19 selloff, it dropped by about 34% from February to March 2020. Our goal was to design a systematic, rules-based trading algorithm that matches or beats the S&P 500's long-run return while materially reducing tail risk and maximum drawdown.

We chose **cross-sectional price momentum** as our signal, one of the most robust and well-documented anomalies in financial research, enhanced with volatility scaling, trend filtering, and a short overlay for hedging.

Project Goals

- Build a fully automated, cloud-deployed trading system that rebalances monthly on Alpaca
- Outperform SPY on a risk-adjusted basis, targeting higher Sharpe and Calmar ratios in both in-sample and out-of-sample testing.
- Reduce maximum drawdown to <30% while capturing upside swings
- Implement four sequential risk controls: volatility scaling, sector caps, trend filter, along with position and liquidity limits
- Deliver a live dashboard with portfolio positions, monthly P&L, and AI-powered Q&A via Groq
- Use real Bloomberg and FRED data for signal generation, portfolio construction, and macro monitoring

1. Risk Management Configs

These controls are applied before any trade is executed:

1.Volatility Scaling

Target 12% annualized vol. Scale factor [0.15-2.0] based on 126-day trailing realized vol. Automatically reduces exposure during turbulent markets.

2.Trend Filter

If SPY trades below its 200-day moving average, we reduce total exposure to 50%. This prevents deploying full capital during bear markets, reducing risk.

3.Sector Caps

No single GICS sector may exceed 30% of either leg. Prevents concentration in a single theme (ex: all-tech momentum crash in 2000).

4.Position & Liquidity Limits

Max position is 5% of portfolio value. Stocks need a minimum $5M avg daily volume (20 day). Prevents over concentration.

5.Drawdown Breaker

Disabled - Creates a doom loop where one severe drawdown locks the strategy at 50% exposure indefinitely, preventing recovery.

6.Macro Regime Filter

Disabled - Backtesting showed false positives in non-crisis environments, hurting Sharpe out of sample. VIX, yield curve, and credit spread are available.

2. AWS Cloud Architecture & Deployment

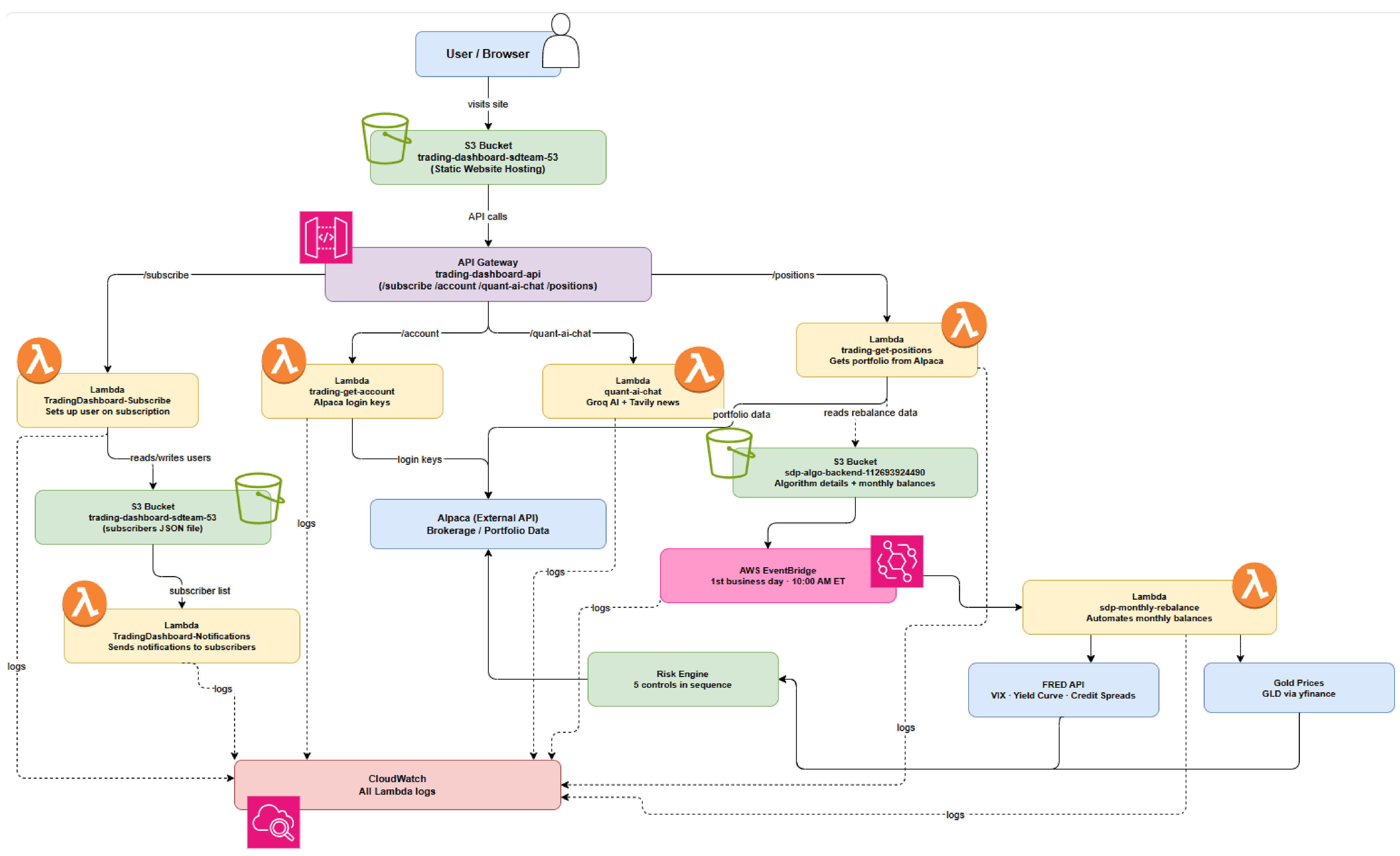

Figure 1 - Our system is deployed on a fully automated AWS pipeline. We utilize S3 to host the interactive dashboard and API Gateway to route requests to Lambda functions for data ingestion. For complex calculations, EC2 handles heavy compute tasks while S3 buckets serve as a central data lake. EventBridge orchestrates our monthly rebalancing and CloudWatch monitors system health, creating a cost-effective, hybrid architecture that ensures reliable, serverless execution.

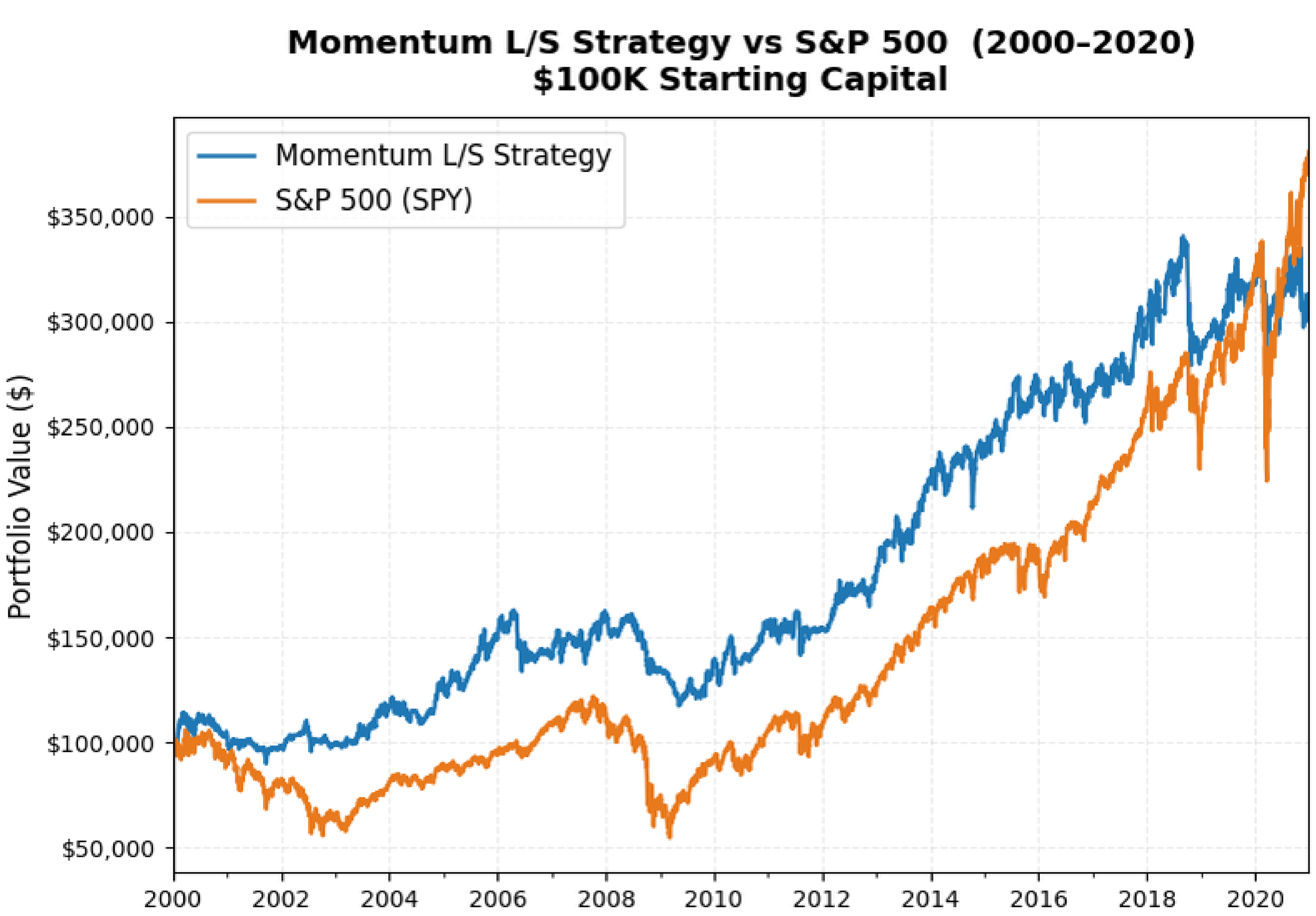

Figure 2 - $100K starting capital | 2000-2020 backtest.

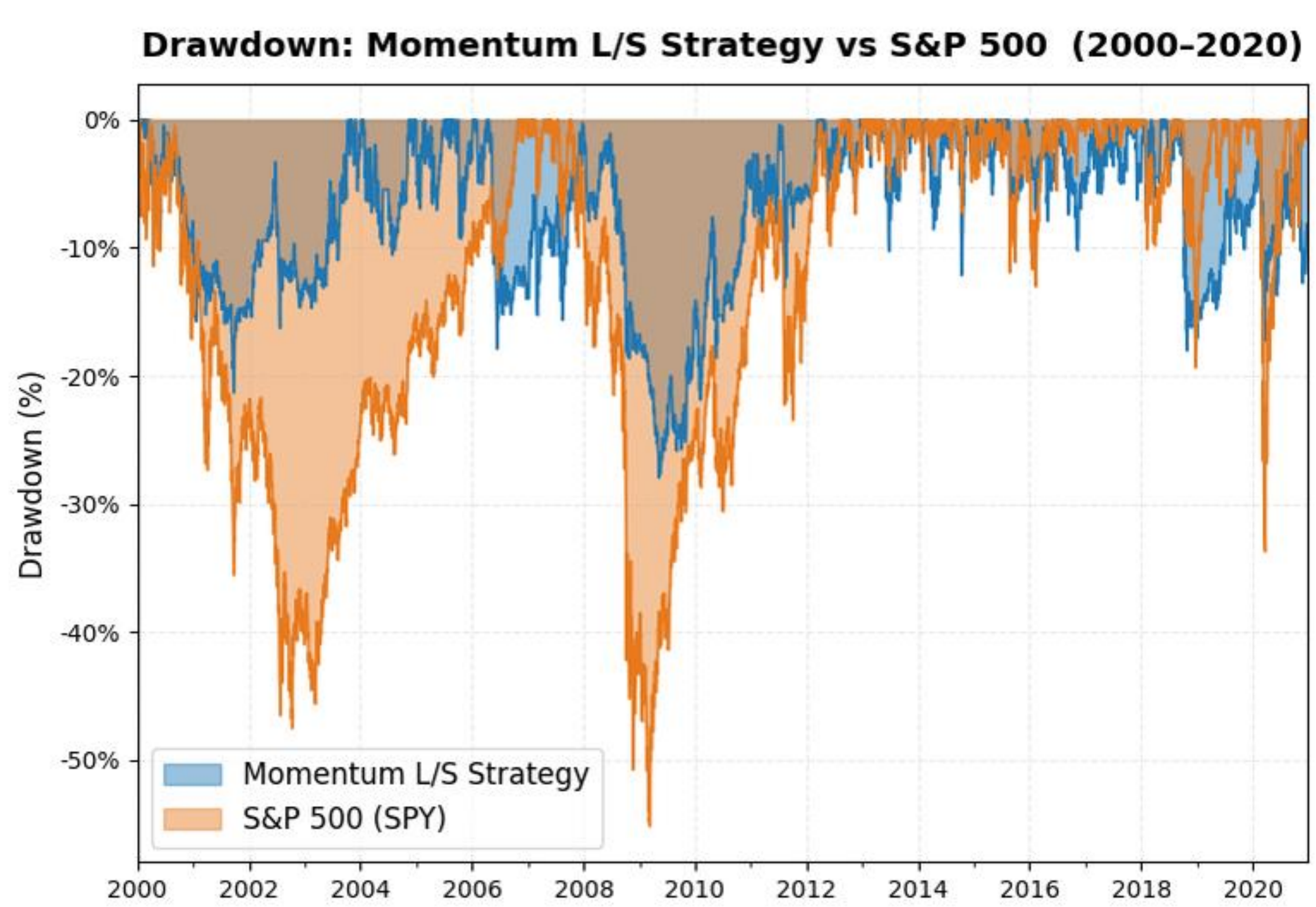

Figure 3 - $100K starting capital | 2000-2020 backtest.
Strategy (blue) substantially shallower than S&P 500 through both the 2008 Financial Crisis and 2020 Covid Crash.

Results

CAGR - Strategy	Sharpe Ratio	Max Drawdown	Calmar Ratio	Volatility
5.56%	**0.47**	**-27.95%**	**0.20**	**13.43%**
vs. SPY 6.58%	vs. SPY 0.42	vs. SPY -55.19%	vs. SPY 0.12	vs. SPY 19.90%